

STMU MIS TASK & HELPDESK PLATFORM

Comprehensive Features, Workflows & Operational Architecture Guide

Version: 2.5 (Enterprise Edition) | Target: Shifa Tameer-e-Millat University MIS Directorate | Date: October 2026

1. Executive Summary & Architecture Overview

The **STMU MIS Task & Helpdesk Service Platform** is an integrated operational platform tailored for the demands of higher education management, clinical hospital systems, and institutional development sprints. It eliminates organizational silos by directly unifying **Agile Sprint Delivery (ClickUp-style)** with **Customer Support Helpdesk Ticketing (Jira Service Desk-style)**.

Layer	Technology Stack & Version	Key Role & Responsibility
Backend Framework	Laravel 12 / PHP 8.3	Robust Eloquent ORM, scheduled jobs, multi-guard auth, REST APIs
Dynamic Frontend	Livewire 3 + Alpine.js	Zero-refresh reactive UI, drawer slide-overs, instant modals
UI Design System	Tailwind CSS v4 + Flux UI	Electric Iris & Obsidian dark mode, micro-animations, glass panels
Data Visualization	Chart.js 4 (Bundled Vite ESM)	Velocity curves, category doughnut, workload matrix, 1000ms easing
AI Intelligence	Google Gemini Multi-Model Pool	Failover pool (Flash-Lite / Flash-Preview), standup & trend synthesis
Biometrics & Security	WebAuthn / FIDO2 Passkeys	Hardware token & fingerprint authentication, 2FA, ISO 27001 logs

2. Hierarchical Organization & Data Model

The system enforces a five-tier hierarchy providing strict multi-tenant containment while allowing cross-departmental coordination:

[University Institution]

- Workspace: MIS Innovation Hub (Multi-Tenant boundary, Slug: /mis)
- Spaces: Core Systems | Hospital IT | Admissions & Student Affairs | ORIC Research
- ■■■ Projects: Student Portal Modernization | Cloud Migration 3.0 | LMS Engine
- ■■■ Task Lists: Sprint Backlog | In Development | QA Verification | Done
- ■ ■■■ Tasks: Checklists, assignees, priorities, due dates, activity logs
- ■■■ Linked Support Tickets: Defect TCK-W1-1049 linked directly to Sprint Project
- Teams & Squads: Core DevOps, ERP Squad (Tracks Lead, Max Capacity & Active Ticket Load)

3. End-to-End Operational Workflows

Workflow 1: Agile Task Lifecycle & Sprint Execution

- Phase A (Inception):** Tasks are initiated within a specific Project and Task List. Attributes include Title, Markdown specs, Priority (Low, Normal, High, Urgent), start date, deadline, assignees, and nested checklist items.
- Phase B (Multi-View Interaction):** Engineers manage work across 4 view modes: (1) *Kanban Board* with drag-and-drop column transitions; (2) *Table View* with inline dropdown adjustments; (3) *Calendar Grid* for milestone deadlines; (4) *Gantt/Timeline* for duration schedules.
- Phase C (Audit Logging):** Every status modification, assignee assignment, checklist check, or comment writes an immutable entry to task_activities with old vs new value snapshots.

Workflow 2: Helpdesk Service Desk, Smart Triage & SLA Resolution

Phase 1 (Submission): Students, faculty, or hospital departments submit tickets with Category, Project, Description, and Attachments. System assigns a formatted serial number (e.g. TCK-W1-4092-49).

Phase 2 (Dynamic SLA Clock): Real-time countdown clock begins ticking based on priority: **Urgent (4h SLA)**, **High (24h SLA)**, **Normal (72h SLA)**, **Low (120h SLA)**.

Phase 3 (Smart Capacity Balancing): TicketCapacityService evaluates agent loads in the assigned team. If an agent has reached their capacity_limit (e.g. 5/5 tickets), an amber/rose overload warning triggers, and the system recommends reassigning to the least-burdened engineer.

Phase 4 (Mandatory Resolution Summary Gate): An engineer cannot mark a ticket Resolved without submitting an exhaustive technical resolution summary detailing the root cause and verification steps.

Phase 5 (Client Rating & Self-Service Reopening): Requesters view their ticket in the zero-trust Requester Portal (/tickets/my), can submit 1-5 star ratings, or reopen within 7 days if the defect persists.

Workflow 3: Interactive Dashboard & Comparative Analytics Hub

1. Granular Time-Travel Selector: Livewire toolbar toggles between **Month-Wise** (day-stepped), **Last Month**, **Year-Wise**, and **Trailing 12-Months** with dynamic Year (2024–2027) and Month (1–12) dropdowns.

2. Three Animated Chart.js Canvases:

- *Velocity Curve:* Smooth bezier lines tracking Tickets Influx, Tickets Resolved, and Tasks Shipped with radiant gradient fills.
- *Category Doughnut:* Issue breakdown (LMS, ERP, Network, Exam Records) with 12px hover offset and center ticket counter.
- *Project Workload Matrix:* Grouped bar chart comparing active task backlogs against incoming defect tickets.

3. Automated MoM Delta Calculations: Automated mathematical comparison calculates net difference and percentage delta: $\Delta\% = ((\text{Current} - \text{Previous}) / \text{Previous}) * 100$.

4. Platform Risk Radar: Instant flags on Overdue Tasks, SLA Breaches, and Blocked Blocker items.

Workflow 4: Google Gemini AI Operations Suite & Zero-Leakage RBAC

High-Availability Multi-Model Failover: Dispatches requests across a tiered pool (gemini-flash-lite-latest → gemini-3-flash-preview → gemini-3.8-flash) with deterministic local mathematical fallbacks.

Key Operational Capabilities:

- *Executive Team Briefing:* High-level throughput, blocker bottlenecks, and sprint velocity synthesized for Directors.
- *Personal Daily Standup:* Automatically formats accomplishments, today's targets, and obstacles for daily Scrum.
- *Trend Diagnostics:* Answers ad-hoc natural language prompts (e.g. 'Why did tickets spike this week?').
- *Zero-Leakage Security:* Server-side prompt assembly filters strictly by user permissions, preventing cross-tenant or private record leaks.

Workflow 5: Biometric Passkeys (WebAuthn) & ISO 27001 Audit Trail

Passwordless Authentication: Users register FIDO2 WebAuthn credentials (Apple TouchID, FaceID, Windows Hello, YubiKeys). Public keys are stored cryptographically; zero plaintext passwords traverse the network.

Compliance Telemetry: Chronological activity log (/activity-logs) records every login, task transition, ticket assignment, and data export, providing full CSV exportability for ISO 27001 accreditation.

4. Comprehensive Module Feature Specification Matrix

Module	Route / Endpoint	Core Capabilities	Underlying Models / Services
Dashboard Hub	/workspace/{slug}	Playable velocity curves, category doughnut, workload matrix, MoM delta math, time-travel toolbar, risk radar.	Dashboard.php, Chart.js, Ticket, Task, Project
Kanban Board	/workspace/{slug}/tasks	Drag-and-drop columns, priority badges, checklist chips, task detail drawer, custom status creation.	TaskManager.php, Task, TaskStatus, TaskList
My Tasks	/workspace/{slug}/tasks/my	Personal focus queue segmented by Overdue, Due Today, This Week, and Completed sprints.	MyTasks.php, Task, User
Triage Queue	/workspace/{slug}/tickets/queue	Incoming ticket triage, live SLA countdown timers, priority filters, project association, engineer assignment.	TicketQueue.php, Ticket, Project, Category
Agent Capacity	/workspace/{slug}/tickets/capacity	Workload heatmaps, capacity limit thresholds (e.g. 5 tickets), rebalancing recommendations.	TicketCapacityService.php, Team, User
Requester Portal	/workspace/{slug}/tickets/my	Zero-trust client portal, ticket creation, real-time tracking, 5-star rating, self-service reopen.	MyTickets.php, Ticket
Gemini AI Suite	Header / AI Bar	Executive summaries, agile standups, natural language trend diagnostics, multi-model failover pool.	GeminiService.php, Google Gemini API
Audit Telemetry	/workspace/{slug}/activity-logs	Chronological immutable event trail, user/type filtering, ISO 27001 compliance, CSV data export.	ActivityLogs.php, TicketActivityLog, TaskActivity
Passkeys & 2FA	/settings/security	FIDO2 WebAuthn biometric passkeys, TOTP two-factor codes, recovery codes, session revocation.	PasskeyController.php, laravel/passkeys
CSV Exports	/workspace/{slug}/export	Full database export of tasks, tickets, SLA metrics, and compliance logs in universal CSV format.	ExportController.php, Storage

5. User Roles, Security Boundaries & Personas

Persona / Name	Role	System Privileges & Boundaries	Authorized Areas
Muhammad Ali muhammad.ali@stmu.edu.pk	Workspace Owner	Complete administrative authority, workspace billing, space creation, telemetry export, AI digests.	All Platform Views & System Settings
Khubaib Ur Rehman khubaib@stmu.edu.pk	Admin / Team Lead	Triage queue management, sprint planning, ticket assignment, agent capacity rebalancing, standups.	Workspaces, Tasks, Triage, Teams, Capacity
Hamza / Sarah / Khurram Engineering Staff	Staff Member	Sprint execution, checklist progress, defect resolution, mandatory resolution summary documentation.	Task Boards, Personal Tasks, Assigned Tickets
Sara Khan (Faculty/Student) requester@example.com	Requester	Zero-trust restricted client. Can ONLY view and submit own support tickets. Cannot access internal sprint tasks.	Exclusively restricted to /tickets/my
Faisal Kamran (ISO Auditor) Auditor Guest	Guest / Auditor	Read-only verification of SLA compliance rates, resolution timestamps, and immutable audit logs.	Audit Activity Logs & CSV Exports

6. Quality Assurance & Verification Standards

The platform includes automated PHPUnit / Pest integration suites certifying data integrity, SLA math, and Livewire reactivity:

- Dashboard Interactive Suite:** DashboardChartsInteractiveTest.php certifies time-travel filters, dataset counts, and Chart.js DOM bindings.
- End-to-End Integration Suite:** TicketingAndTasksIntegrationTest.php certifies full cross-functional workflows from ticket intake to resolution and task completion.
- Test Execution Status:** All 9 integration tests passing (100% assertions verified).

Authored by STMU MIS Directorate & Advanced Engineering Team • Built with Laravel, Livewire, Chart.js & Google Gemini AI.